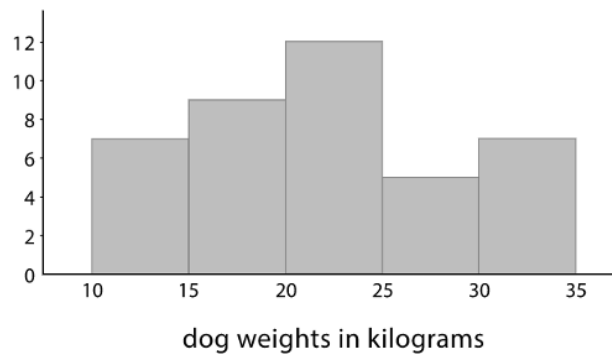
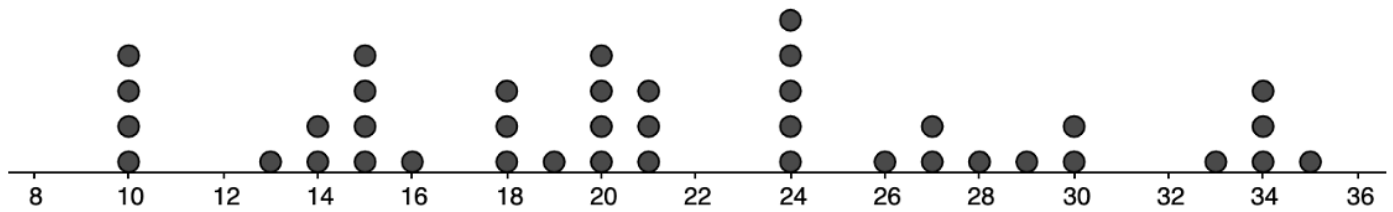


Grade 6 Accelerated
Unit 9
Lesson 8 Homework

Name Answer Key
Date _____
Period _____

A histogram and dot plot show the weights, in pounds, of 40 dogs at a dog show.



1. Which data display can be used to determine the mean weight?

Dot plot because you can see actual data

2. Find the mean weight. Interpret the mean.

$$\frac{860}{40} = 21.5$$

The average weight of the dogs was 21.5 lbs.

3. Which data display can be used to determine the median weight?

Dot plot

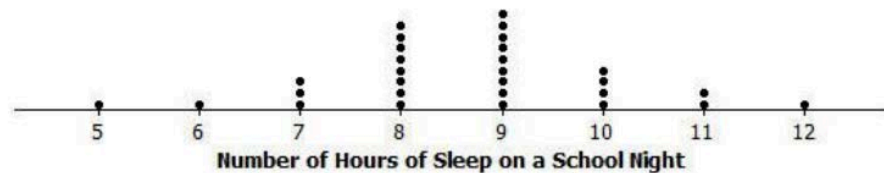
4. Find the median weight. Interpret the median.

$$\frac{20+21}{2} = 20.5$$

The middle value of the dogs' weights was 20.5 lbs. Half of the dogs weighed more than 20.5 lbs., and half weighed less than 20.5 lbs.

Rocky is a sixth grader at Roosevelt Middle School and investigated the number of hours of sleep sixth-grade students get on school nights. He surveyed 29 sixth graders and organized his data on the dot plot.

Dot Plot of Number of Hours of Sleep



5. Describe the spread and shape of the distribution.

Symmetrical with a range of 7 hours

6. Find the mean number of hours slept by 6th graders.

$$\frac{251}{29} = 8.7 \text{ hours}$$

7. Interpret the mean.

The average amount of hours a sixth grade student sleeps is 8.7 hours.

8. Find the median.

9

9. Interpret the median.

The middle value of the dataset is 9 hours.

10. Suppose Rocky surveyed 10 more students and they reported the following (in hours); 9, 10, 9, 9, 12, 12, 12, 8, 10, 7. If these 10 data points were added to the dot plot, explain how would the mean and median change?

$$\text{Mean: } \frac{34.9}{39} \approx 8.9 \quad \text{Median still} = 9$$

With more data, the mean and median come closer to being the same value.